O-C-240d
March 25, 1963

SUPERSEDING
Int. Fed. Spec. O-C-00240c (GSA-FSS)
May 27, 1960 and
Fed. Spec. O-C-240b
May 7, 1957

FEDERAL SPECIFICATION

CHARGE, FIRE EXTINGUISHER, ANTIFREEZE AND CHARGE, FIRE EXTINGUISHER, SODA-ACID

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers charges for water (including pump tank and gas-expelled liquid) and soda-and-acid fire extinguishers. Type I charges are intended for use in winterizing water-type fire extinguishers, and will prevent the solution from freezing at a temperature of minus 40°F. Type II charges are intended for use in standard soda-and-acid-type fire extinguishers.

1.1.1 *Federal specification coverage.* Federal specifications do not include all types and classes of the commodity indicated by the titles of the specifications nor all those which are commercially available, but are intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* The fire extinguisher charges covered by this specification shall be of the following types and classes, as specified (see 6.1):

Type I.—Antifreeze charge for water-type extinguishers.
- Class A.—For 2-1/2-gallon extinguishers.
- Class B.—For 4-gallon extinguishers.
- Class C.—For 5-gallon extinguishers.
- Class D.—For 40-gallon extinguishers (on wheels).

Type II.—Charge for soda-and-acid extinguishers.
- Class A.—For 2-1/2-gallon extinguishers.
- Class B.—For 40-gallon extinguishers (on wheels).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

O-C-105—Calcium Chloride, Dehydrate and Calcium Chloride Anhydrous; Technical.
O-E-927—Extinguisher, Fire, Soda-And-Acid, (Hand).
O-S-576—Sodium Bicarbonate; Technical Grade.
O-S-588—Sodium Chromate, Anhydrous, Technical.

FSC 4210

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

O-S-809—Sulfuric Acid, Technical.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-C-96—Cans, Metal, 28 Gage and Lighter.
PPP-D-723—Drums, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

JAN-D-195—Drums, Steel, Calcium Carbide (100-Pound Capacity).

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Code of Federal Regulations:

49 CFR 71-78—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition—Revised 1956) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders should cite "49-CFR 71-78".)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Unless otherwise specified, quantities of materials expressed herein are on the avoirdupois weight basis.

3.1.1 *Calcium chloride dihydrate.* Calcium chloride dihydrate shall conform to the requirements of O-C-105, type I, class 2.

3.1.2 *Sodium bicarbonate.* Sodium bicarbonate shall conform to the requirements of O-S-576.

3.1.2.1 *Noncaking additive.* When specified in the invitation to bid (see 6.1), a maximum of 0.75 percent of calcium hydroxyphosphate may be uniformly mixed with the sodium bicarbonate to effect noncaking and free-flowing. The mixture need not be a clear solution as required by the solubility test of O-S-576 but shall go into solution upon shaking and shall comply with the requirements for loss on heating, chlorides, and total alkalinity as specified in O-S-576.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

3.1.3 *Sulfuric acid.* Sulfuric acid shall conform to the requirements of O-S-809, type I, class 2.

3.1.4 *Anhydrous sodium chromate.* Anhydrous sodium chromate shall conform to the requirements of O-S-588.

3.2 **Type I, charge for water-type extinguishers.** Type I charge shall be used only in extinguishers marked to permit the use of water and antifreeze solution. In no case shall type I charge be used in stainless steel extinguishers.

3.2.1 *Class A.* Class A charge (for 2-1/2-gallon extinguisher) shall consist of 10 pounds, plus or minus 4 ounces, of calcium chloride dihydrate together with one ounce (minimum) of anhydrous sodium chromate (see 4.4).

3.2.2 *Class B.* Class B charge (for 4-gallon extinguisher) shall consist of 16 pounds, plus or minus 4 ounces, of calcium chloride dihydrate together with 1-1/2 ounces (minimum) of anhydrous sodium chromate (see 4.4).

3.2.3 *Class C.* Class C charge (for 5-gallon extinguisher) shall consist of 20 pounds, plus or minus 4 ounces, of calcium chloride dihydrate together with 2 ounces (minimum) of anhydrous sodium chromate (see 4.4).

3.2.4 *Class D.* Class D charge (for 40-gallon extinguisher) shall consist of 132 pounds, plus or minus one pound, of calcium chloride dihydrate together with 12 ounces (minimum) of anhydrous sodium chromate (see 4.4).

3.3 **Type II.** Type II charges shall consist of acid and soda components.

3.3.1 *Class A (for 2-1/2-gallon-extinguisher).*

3.3.1.1 *Acid component.* The acid component shall consist of 7-1/2 ounces, plus or minus 1/2 ounce, (4 fluid ounces plus or minus 1/4-fluid ounce) of sulfuric acid. The acid shall be furnished in an 8-fluid ounce, heat-resistant bottle conforming to the requirements of O-E-927.

3.3.1.2 *Soda component.* The soda component shall consist of 1-1/2 pounds, plus or minus 1 ounce, of sodium bicarbonate.

3.3.2 *Class B (for 40-gallon extinguisher).*

3.3.2.1 *Acid component.* The acid component shall consist of 7 pounds, plus or minus 1/2 ounce, (58.8-fluid ounces plus or minus 1/4-fluid ounce) of sulfuric acid.

3.3.2.2 *Soda component.* The soda component shall consist of 20 pounds, plus or minus one ounce, of sodium bicarbonate.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling.** Sampling for visual inspection shall be in accordance with MIL-STD-105.

4.3 **Inspection.** Each sample shall be visually inspected for broken, damaged, or improper container; improper closure; leakage; improper or illegible marking. Samples containing one or more defects shall be re-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

jected. Rejection of lots shall be based on an AQL of 2.5 percent defective.

4.4 Test for sodium chromate (for type I charge for water-type extinguisher).

4.4.1 *Preparation of standard solution.* The standard solution shall be prepared as follows: Weigh exactly 3 grams of anhydrous sodium chromate conforming to 3.1.4 on an analytical balance and approximately 480 grams of calcium chloride dihydrate conforming to 3.1.1 on a rough balance. Dissolve the weighed salts in a 1000-ml. beaker with approximately 600 ml. of distilled water. Mix thoroughly and make up to volume in a 1000-ml. volumetric flask, according to standard practice. Pipette a 1-ml. aliquot of this standard solution into a clean 100-ml. Nessler tube. Add 5 ml. of concentrated hydrochloric acid and dilute exactly to the mark with distilled water. Place a rubber stopper in top of tube and invert several times to mix. Place the tube with contents in the Nessler tube rack for color comparison with the test sample.

4.4.2 *Preparation of test sample.* The test sample shall be prepared as follows: Dissolve a selected whole fire extinguisher charge in the amount of distilled water as required by the charge make-up directions and thoroughly mix. Take an approximately 500-ml. sample, according to best sampling technique, using a glass sampling tube. Pipette 1 ml. of this solution into a clean 100-ml. Nessler tube. Add 5 ml. of concentrated hydrochloric acid and dilute exactly to the mark with distilled water. Place a rubber stopper in top of tube and invert several times to mix.

4.4.3 *Test procedure.* The color intensity of the two Nessler tubes, one containing the standard solution equivalent to one ounce of sodium chromate per 2-1/2-gallon charge, and the other containing the test sample charge, shall be compared side by side in the rack by viewing through the liquid column under a satisfactory light. If the color intensity of the test sample is equal to or greater than that of the standard, the test charge shall be considered as having the proper amount of sodium chromate. If the color intensity is less than that of the standard, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

(**Civil agencies.** Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 General.

5.1.1 *Type II soda-and-acid charges, classes A and B.*

5.1.1.2 *Acid component.* The class B acid component shall be packaged in bottles commonly used for the purpose, which shall have sufficient outage allowance to provide for the expansion of the contents from change of temperature. Class A bottle shall conform to requirements of O-E-927 (see 3.3.1.1). The bottles for both the class A (see 3.3.1.1) and class B components shall be furnished with either metal or plastic continuous-thread screw caps. Caps shall be fitted with sulphuric acid-resistant liner and tightened to prevent leakage.

5.2 Packaging.

5.2.1 *Level A.*

5.2.1.1 *Type I, antifreeze charges.* Classes A, B, and C charges shall be unit packaged in multiple-friction top, metal cans conforming to PPP-C-96, type V, class 2. The plugs (lids) shall be closed and secured in conformance with the can specification. Class D charges shall be packaged in steel drums conforming to JAN-D-195, or in metal drums of similar design and construction.

5.2.1.2 *Type II, soda-and-acid charges, classes A and B.*

5.2.1.2.1 *Soda-and-acid components.*

5.2.1.2.1.1 *Class A charge.* The soda com-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

ponent for the class A charge shall be packaged in commercial fiberboard boxes. Each box shall be of a size approximately 3 by 3 by 7 inches and shall contain the soda component and the bottle containing the acid component, comprising one complete charge. Complete directions for recharging the fire extinguishers and a caution notice that the box contains poison shall be printed on the outside of each box.

5.2.1.2.1.2 *Class B charge.* The soda for the class B charge shall be packaged in fiber drums conforming to PPP-D-723, type I. A bottle containing the acid component shall be placed in the center of the respective fiber can and drum, completely surrounded with the sodium bicarbonate. The bottle containing the acid component for the class B charge shall be suspended in the center of the fiber drum by means of adequate supports and the bottle surrounded by the soda component which will neutralize the acid in the event of bottle breakage.

5.2.2 *Level C.*

5.2.2.1 The components of each charge (except type II, class A) shall be packaged to afford protection against deterioration and damage in conformance with commercial practice. Type II, class A sulphuric acid component shall be packed in a glass bottle (see 5.1.1.2) and the bottle placed in a container (conforming to ICC regulations) and completely surrounded with the sodium bicarbonate.

5.3 **Packing.**

5.3.1 *Level A.*

5.3.1.1 Type I, class D charges will require no overpacking. Type I, classes A, B, and C, and type II, class B charges shall be packed in quantities not to exceed 200 pounds gross weight, in snug-fitting nailed plywood boxes conforming to PPP-B-621, or wood-cleated plywood boxes conforming to PPP-B-601 (overseas type). The boxes shall be closed and strapped in conformance with the appendix to the applicable box specification. Type II, class A charges shall be packed in snug-fitting fiberboard boxes conforming to PPP-B-636, V3c or V3s, class 2.

5.3.2 *Level B.*

5.3.2.1 Type I, class D charges will require no overpacking. Type I, classes A, B, and C, and type II, class B charges shall be packed in quantities not to exceed 200 pounds gross weight in snug-fitting boxes conforming to PPP-B-621, class 1; PPP-B-601, domestic type; or PPP-B-585, class 1. Type II, class A charges shall be packed in snug-fitting boxes conforming to PPP-B-636, class 1.

5.3.3 *Level C.*

5.3.3.1 The packaged charges shall be packed for shipment in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.4 **Marking.** When applicable, marking of containers shall conform to Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles. All type I charges shall bear the following warning:

WARNING: Harmful dust. May cause rash or external ulcers.
Avoid contact with skin and eyes.
Avoid inhaling dust or solution spray.

5.4.1 *Civil agencies.* In addition to any special marking required, unless otherwise specified, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 *Military.* In addition to any spe-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

cial marking required, marking for identification, shipment, and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of charge required (see 1.2.1).
(*c*) Specify if noncaking additive is to be mixed with the sodium bicarbonate (see 3.1.2.1).
(*d*) State levels of packaging and packing required (see 5.2 and 5.3).
(*e*) Marking, if different (see 5.4).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Fire extinguisher charges, not otherwise indexed by name.
Carload minimum weight 36,000 pounds.

Motor:

Fire extinguisher charges, not otherwise indexed.
Truckload minimum weight (W) 36.6 pounds, (W) subject to Rule 115, National Motor Freight Classification.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—E
Navy—Y
Air Force—Wright Air Development Division

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.